

On-board state of health estimation of LiFePO₄ battery pack through differential voltage analysis



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HIGHLIGHTS

- An on-board SOH monitoring algorithm is proposed by using partially data.
- Improved center least squares method is used to obtain the differential curves.
- Two inflection points location interval is used to represent battery SOH.
- First inflection point location can eliminate the charged capacity cumulative error.
- Establish a quantitative correlation between transformation parameter and battery SOH.

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ABSTRACT

Battery state of health (SOH) estimation is a crucial challenge for electric vehicles due to the complicated aging mechanism of a battery. In this paper, a new method for battery SOH estimation is proposed using the location interval between two inflection points or the transformation parameter of the differential voltage (DV) curve. And a new algorithm to obtain the DV curve according to the center least squares method is presented. Four LiFePO₄ battery cells in different aging states are tested to validate the proposed method. Results show that the linear regression of the location interval between two inflection points versus the battery capacity from one single battery cell is able to evaluate the SOH of other three battery cells within 2% error bound. Transforming the measured DV curve to be congruent with the initial DV curve provided by the manufacturer, battery cell SOH can be estimated according to the transformation parameter with the error being within 2.5%. Furthermore, a battery pack SOH monitoring framework is inferred from the inflection point location and the transformation parameter of the DV curve to realize battery pack SOH estimation.

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1. Introduction

The growing concern over fuel consumption and carbon emissions has greatly promoted the development of electric vehicles (EVs) [1]. The battery pack is an essential part of an EV and its performance directly affects the vehicle range, acceleration performance and charging acceptance during energy recovery from regenerative braking [2,3].

In an EV, the battery pack should be monitored, controlled and managed by a battery management system (BMS) to keep it working properly. For a BMS, the core tasks are the state of charge (SOC) estimation and the state of health (SOH) determination [4–6].

Lu [7] summarized and analyzed the commonly used algorithms for SOC estimation. Generally, these are the standard current integration method combined with the open circuit voltage method [8], or the Kalman filtering method [9,10], or the sliding mode observer method [11]. And the estimation error can be less than 5% under some condition.

SOH estimation is more challengeable due to the complicated aging mechanism of a battery. Typically, the quantitative definition of SOH is based on capacity decreasing or internal resistance increasing. The multi-scale extended Kalman filtering method is considered as the most promising method for on-line SOH estimation [12,13]. The state equation for SOH quantitative description is added to the traditional extended Kalman filtering method. The SOC and SOH are then computationally solved at different time intervals. However, this method is highly depended on the adopted battery model and the accuracy of the relationship between

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battery open circuit voltage and SOC. In addition, the parameters of a battery model are influenced by the battery aging, operation mode and work environment. Therefore, the SOH estimation accuracy relies not only on the reliability of the battery model, but also on the accuracy of the model's determined parameters [14].

Recently, many researchers have concentrated on the study of the lithium intercalation process and the corresponding staging phenomenon of a battery [15–17]. The battery charge voltage curve (charge voltage vs. charged capacity, $V-Q$) can be transformed to form the differential voltage ($DV, \frac{dV}{dQ} - Q$) curve or the incremental capacity ($IC, \frac{dQ}{dV} - V$) curve. The battery aging state can then be described by the peaks in the DV or IC curve. Groot et al. [17–19] evaluated the battery aging state according to the DV curve. This method is called the differential voltage analysis (DVA) method. Accordingly, Smith et al. [3,20–22] estimated the battery aging state by the incremental capacity analysis (ICA) method. In contrast to the DVA, the ICA is based on the IC curve. Although the DVA or ICA method is an effective method for describing battery aging process, the two methods are still remain in the electrochemical aging mechanism research, and not in on-board application. In addition, because of battery cell inconsistency in a series-connected battery pack, some battery cells cannot reach the peaks in DVA or ICA method.

As the measured voltage contains noise or trembling, it is inevitable that some perturbation is introduced into the DV or IC curve [23]. So it is difficult to identify the peak that appeared in the voltage plateau region of the DV or IC curve automatically. Weng [3] chose a 5th order polynomial to fit the measured voltage curve, and calculated the IC curve from that fitted curve. Although the polynomial curve fitting gives a much smoother IC curve, the method is constrained by the order of the polynomial and the used data region. A more robust method for calculating the DV or IC curve is desirable.

In this paper, four LiFePO₄ battery cells in different aging states are examined to acquire measured charge voltage curves. An improved center least squares method is introduced and used to calculate the DV curve. A new algorithm is then proposed to predict the battery capacity aging according to the location interval between the two inflection points or the transformation parameter of the DV curve.

2. Battery cycle test and data analysis

2.1. Battery testing system and schedule

Four LiFePO₄ battery cells in different aging states are examined to acquire charge voltage curve. The battery cells are labeled as No. 1–No. 4 battery cell. The specifications of equipments and the parameters of the battery cells used in the tests are list in Table 1. Before life-cycle testing, each battery cell's capacity is measured under environmental temperature (25 °C) according to the "2010 EV lithium-ion power battery test specification". This specifies a full charged battery cell is discharged at 1 C rate until the battery voltage falls to the low voltage limit and then its capacity is measured, C being the rated capacity in ampere-hours. The original

measured capacity of No. 1–No. 4 battery cell is 4.1 Ah, 4.08 Ah, 3.47 Ah and 3.51 Ah.

The four battery cells are life cycle tested on a cycling regime consisting of charging at the 1 C rate and discharging at the 1 C rate at environmental temperature. In each cycle, as shown in Fig. 1, all the battery cells are charged at the 1 C rate to 3.8 V and followed by a constant-voltage phase until the charge current lows into 0.01 C. The battery cells are then discharged at 1 C rate to 2 V. After discharging, the battery cells are given a 2 h rest before recharging. The cycle numbers performed for the four battery cells are about 1549, 1517, 363 and 451.

2.2. Cycle charge voltage curve analysis

Battery discharge current changes drastically during vehicle driving conditions. However, the charge current is constant under most conditions. Therefore, the charge $V-Q$ curve is chosen to investigate the method for describing the battery aging state.

Fig. 2 shows the battery charge voltage curves of No. 1 battery cell under different cycle numbers. Fig. 2(a) gives the battery charge voltage-to-time curves and Fig. 2(b) displays the charge voltage-to-charged capacity curves. The charged capacity in Fig. 2(b) is calculated by the standard current integration method. As shown in the figures, the charge voltage changes slightly in the voltage platform stage under different cycle numbers. It is well known that the battery terminal voltage is subject to the open circuit voltage, charge current and internal resistance. The lower internal resistance is, the smaller voltage drop across the internal resistance will be. The terminal voltage variation caused by the internal resistance increasing is not obvious, which means that the internal resistance or the terminal voltage in the voltage platform stage is not suitable for estimating the battery aging state under some condition. It can also be seen from the figures that the decrease of the battery capacity brings forward the constant-voltage charging phase of the charging process. When the charging process enters the constant-voltage charging phase, these charge voltage curves can be observed accurately under different cycle numbers.

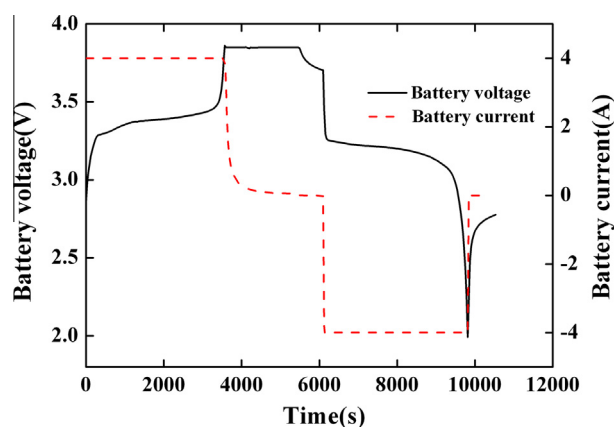


Fig. 1. Battery cell test schedule.

Table 1
Specifications of equipments and parameters of battery cells.

Item	Model	Manufacturer	Characteristics
Constant temperature and humidity test chamber	BE-TH-225L8	DongGuan Bell Experiment Equipment Co., Ltd.	Temperature: −70 °C to 150 °C (±1 °C); humidity: 20–98%
Battery performance tester	HBT-5V10A	NingBo BaTe Technology Co., Ltd.	Voltage: 0–5 V (0.08 FS); current: 150 mA to 10 A (0.08% FS)
LiFePO ₄ battery	GH-4Ah	GuoHeng Energy Co., Ltd.	Capacity: 4 Ah; rated voltage: 3.2 V

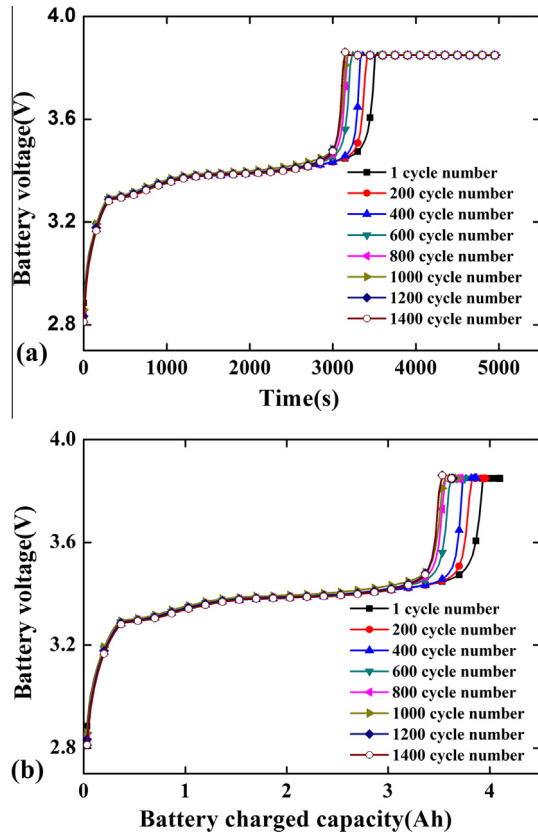


Fig. 2. Battery (a) charge voltage-to-time curves and (b) charge voltage-to-charged capacity curves under different cycle numbers.

3. DV curve acquisition

3.1. Numerical-derivative with smoothing method

The measured data is calculated with numerical-derivative method in order to acquire the differential curve. The measured data can be represented as:

$$y_i = f(x_i) \quad (1)$$

where $x_i = x_0 + i \cdot \Delta x$, $i = 0, 1, 2, \dots$

The derivation value can be calculated as follows:

$$\frac{dy_i}{dx_i} = \frac{f(x_{i+1}) - f(x_i)}{\Delta x} \quad (2)$$

Fig. 3 shows the DV curves of No. 1 battery cell calculated by Eq. (2) at different cycle numbers. The measured voltage is sensitive to the measurement noise, and the calculated DV curve contains this noise. To get a smoother DV curve, the noise DV curve can be processed by the original smoothing method. Fig. 4 shows the corresponding results with a 2nd order polynomial and a moving window containing 20 data points. The inflection point location in the horizontal axis does not shift, while the inflection point values along the vertical axis decreases. Although the smoothed results appear a clear change around the peak region, the results are still very noisy. In addition, the smoothed results are subjected to the selected data range and the order of the polynomial. Thus, a more robust method is desired in the next section.

3.2. Improved center least squares method

If the measured data in a small selected range can be described with a first order polynomial, $y = a_0 + a_1x$, the problem to acquire DV curve is then transformed to solve the parameter a_1 .

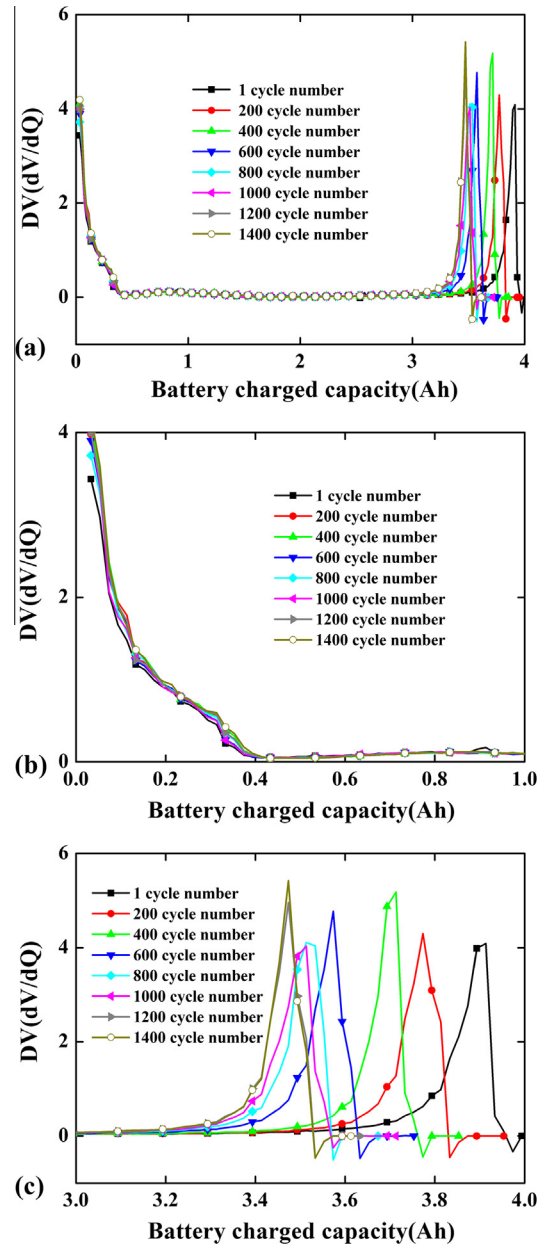


Fig. 3. DV curves with different time sizes of No. 1 battery cell calculated by the numerical-derivative method.

The parameter identification method is as follows, selecting an appropriate length (n) of the measured data, the parameters can be solved by using the constraint

$$Q(a_0^*, a_1^*) = \min Q(a_0, a_1) \quad (3)$$

where $Q = \sum_{k=1}^n (y_k - a_0 - a_1 x_k)^2$.

By solving $\frac{\partial Q}{\partial a_0} = 0$, $\frac{\partial Q}{\partial a_1} = 0$ to minimize the solution of $Q(a_0, a_1)$, then a_0^* , a_1^* can be found.

According to the least squares method, the constraint can be reformulated as:

$$\begin{bmatrix} n & \sum_{k=1}^n x_k \\ \sum_{k=1}^n x_k & \sum_{k=1}^n x_k^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} \sum_{k=1}^n y_k \\ \sum_{k=1}^n x_k y_k \end{bmatrix} \quad (4)$$

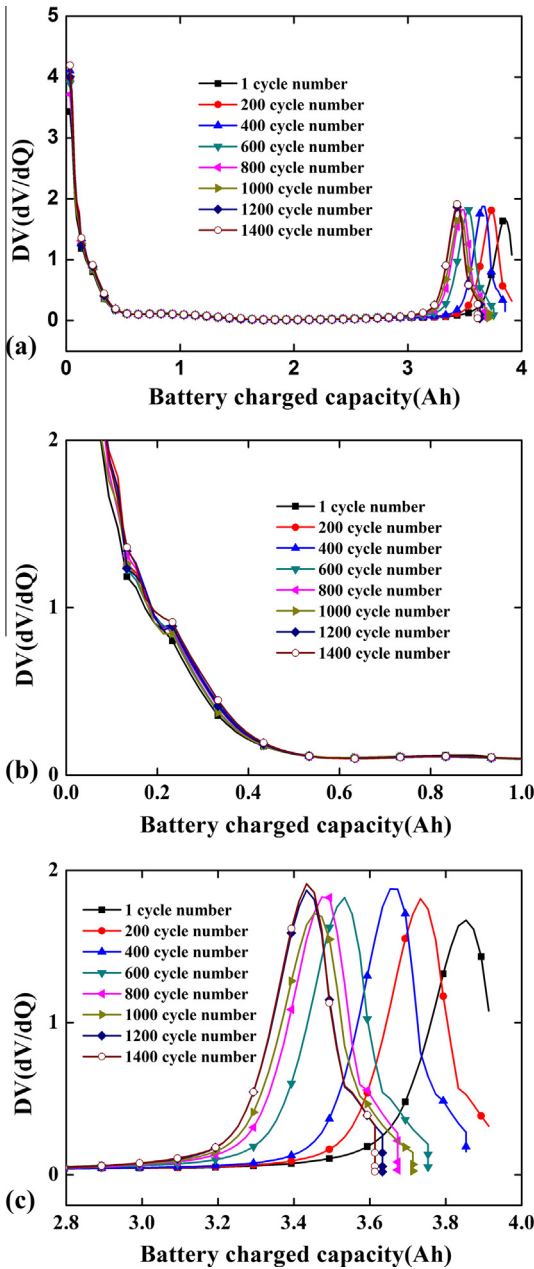


Fig. 4. Smoothed DV curves with different time sizes of No. 1 battery cell with a 2nd order polynomial and a moving window containing 20 data points.

By the improved center least squares method, n data points can be expressed as $(-\frac{n}{2}, y_0), (-\frac{n}{2} + 1, y_1), \dots, (0, y_{\frac{n}{2}}), \dots, (\frac{n}{2}, y_n)$. Then, the parameter a_1 can be solved by:

$$a_1 = \frac{\sum_{k=0}^n (-\frac{n}{2} + k) y_k}{\sum_{k=0}^n (-\frac{n}{2} + k)^2} \quad (5)$$

Fig. 5(a)–(d) show the DV curves obtained by the proposed method with n being 10, 15, 20 and 25. By comparing these figures, it is found that these results have high similarity. However, the DV curves shown in Fig. 5(a) are very noisy and thus the results are not satisfactory. By comparing these results, 15 data points can be chosen in actual application to reduce the computing time. To corresponding with the data point used in the smoothing method, 20 data points is selected in the next discussion. Fig. 6 shows the DV curves obtained by the proposed method with n being 20 used

in the smoothing method. Comparing Fig. 4 with Fig. 6, the inflection points along the horizontal axis in all DV curves remain unchanged. It is also clear that the improved center least squares method gives a smoother DV curve around the inflection point region.

All the DV curves in this paper are calculated using Microsoft Visual C++ 6.0 on a PC with Inter(R) Core(TM) i5-4590 CPU @ 3.30 GHz, RAM 4.0 GB. The computing time of the improved center least squares method and the smoothing method are 15 ms and 17 ms with 1549 cycle number. So, the improved center least squares method will be used in next sections.

4. SOH estimation based on inflection point location in the DV curve

It could be seen from Fig. 6(a) that the first inflection point of the DV curve stays at approximately the same location while the second inflection point moves towards the left with the battery aging. Thus, the second inflection point location or the location interval between the two inflection points in DV curve can be used to evaluate the battery SOH.

Taken No. 1 battery cell's original capacity and its second inflection point location as benchmarks, Fig. 7 shows the correlation between the normalized second inflection point location and the normalized battery cell capacity of the four battery cells. The solid line in the figure gives the fitted linear relationship between the two values of the No. 1 battery cell while the dotted line gives the error bound of $\pm 2\%$. It can be seen from the figure that a strong relationship exists between the second inflection point location and the battery capacity. The deviations between the points and the fitted curve are within 2% with the variance being 1.07, which confirms that the location of the second inflection point can be used to indicate battery capacity fading.

For an EV, the charging current is constant under most conditions. The total charged capacity C can be calculated by the standard current integration method, which can be written as:

$$C = C_0 + It \quad (6)$$

where C_0 represents the initial charged capacity, t is the charged time, I is the charging current.

In order to measure the second inflection point location accurately, it is required to start the charging process from zero SOC, which is difficult in real application. According to Fig. 6(a), the first inflection point in DV curve occurs when the battery charged capacity is about 0.52Ah, and this point is not sensitive to the cycle number. This feature indicates that the first inflection point can be addressed to eliminate the battery capacity cumulative error caused by the standard current integration method. The location interval between the two inflection points can be described as:

$$Q_{DV} = Q_2 - Q_1 \quad (7)$$

where Q_1 and Q_2 are the location of the two inflection points.

The relationship between the normalized Q_{DV} and the normalized battery capacity is shown in Fig. 8. The figure indicates that there also exists a strong relationship between the location interval of the two inflection points and the battery capacity. The deviations between the points and the fitted line are within 2% with the variance being 0.89. The variance is less than that obtained by using the second inflection point location only.

According to the Figs. 7 and 8, the linear regression of the No. 1 battery cell also fit very well with the results from the other three battery cells. For the battery cells from a same manufacturer, the linear relation to describe the relationship between the battery capacity and the location interval of the two inflection points should be established off-line. Then, the SOH of all the battery cells

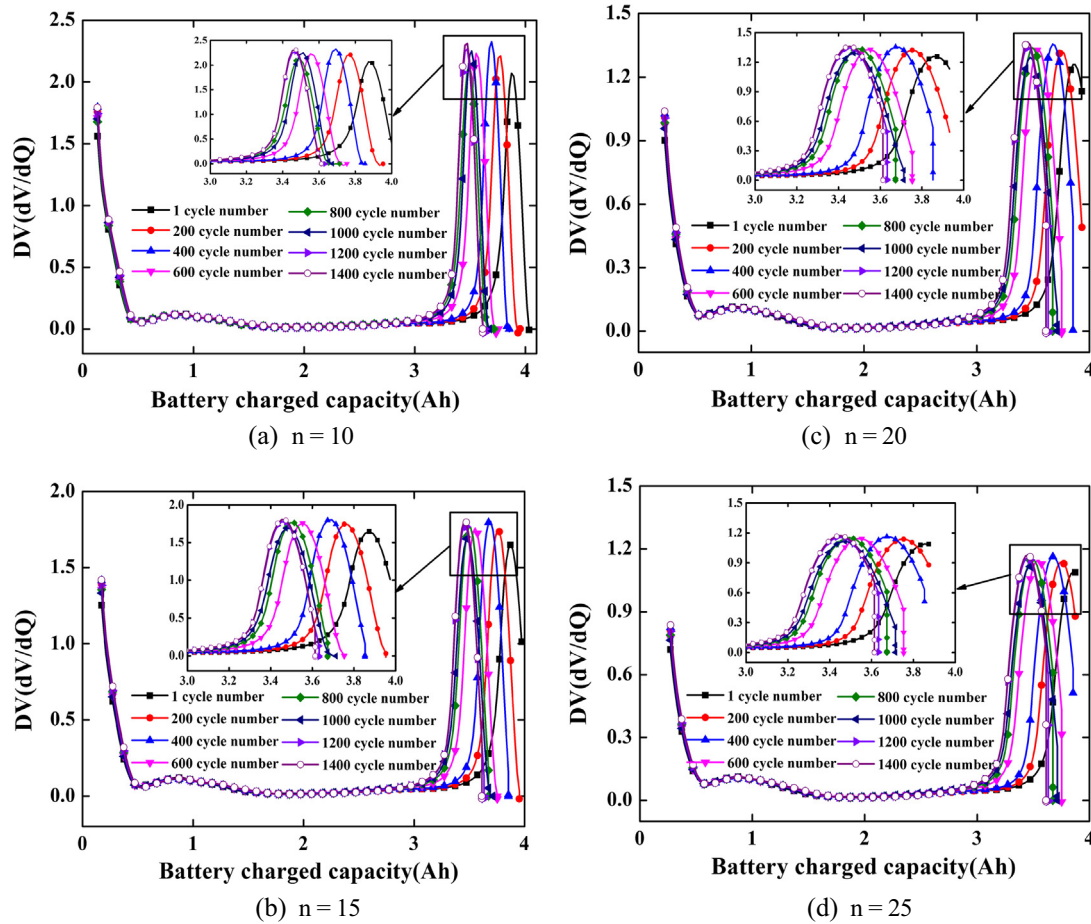


Fig. 5. DV curves of different data point of No. 1 battery cell calculated by the improved center least square method.

in an applied battery pack can be estimated when the location interval of the two inflection points is calculated.

In the charging process, when the battery cell charge curve reaches the first inflection point, the charged capacity calculated by the standard current integration method can be calibrated by a constant value to eliminate the cumulative error. So the first inflection point location can also be used to calibrate the charged capacity.

5. SOH estimation based on transformation parameter of the DV curve

In application, due to the inconsistent manufacturing process and different operating temperature, the inconsistency of battery cell parameters cannot be neglected [24,25]. According to the bucket principle, some battery cells cannot reach the second inflection point when the charging process is completed. Thus, a simple method to estimate the capacity of these battery cells is necessary.

Wang [1] pointed out that the aged voltage-to-capacity curves of the battery cells from the same batch of a manufacturer could be transformed to overlap the original one. Zheng [26] presented a battery capacity estimation method based on charge voltage curve transformation. However, as the objective function was non-linear, this paper also pointed out that the Genetic Algorithm (GA) was still complex in computing the optimum transformation parameter.

Fig. 9(a) is an enlarged view of parts of Fig. 6(a). From the figure, it is clearly evident that the DV curves under different cycle number remain the same till the charged capacity reaches above 2.8 Ah

(SOC is about 70%), which is almost constant for the batteries from the same batch. The moment when appears a significant upward deviation in the DV curve is defined as the separation point. A battery can be served as a power battery till its capacity decayed to 80% of its initial value, which means that each battery cell in a battery pack could reach the separation point at the end of charging process.

From Fig. 9(a), the aged DV curves enter the upward phase earlier with battery aging. So, the difference between the measured DV curve and its original curve which is provided by the manufacturer can be used to indicate the battery aging state.

The difference among the two DV curves can be represented as:

$$G = \sum_{j=1}^m (g_i(Q_j) - g_0(Q_j))^2 \quad (8)$$

where i represents the i -th cycle number, $g_i(Q_j)$ is the measured DV curve of the No. i cycle number, $g_0(Q_j)$ is the original DV curve which is provided by the manufacturer, m is the consecutive selected points in the calculation. The data from the minimum point in the charging voltage flat area, the charged capacity being about 2 Ah, to the end of charging are used for calculating.

The transformation parameter $T_{Qi} = [k_i, Q_{0i}]$ is introduced to convert $g_i(Q_j)$ into $g'_i(Q_j) = g_i(Q_j/k_i + Q_{0i})$ so as to be congruent with $g_0(Q_j)$.

The objective function is constructed as:

$$G = \sum_{j=1}^m (g_i(Q_j/k_i + Q_{0i}) - g_0(Q_j))^2 = G_{\min} \quad (9)$$

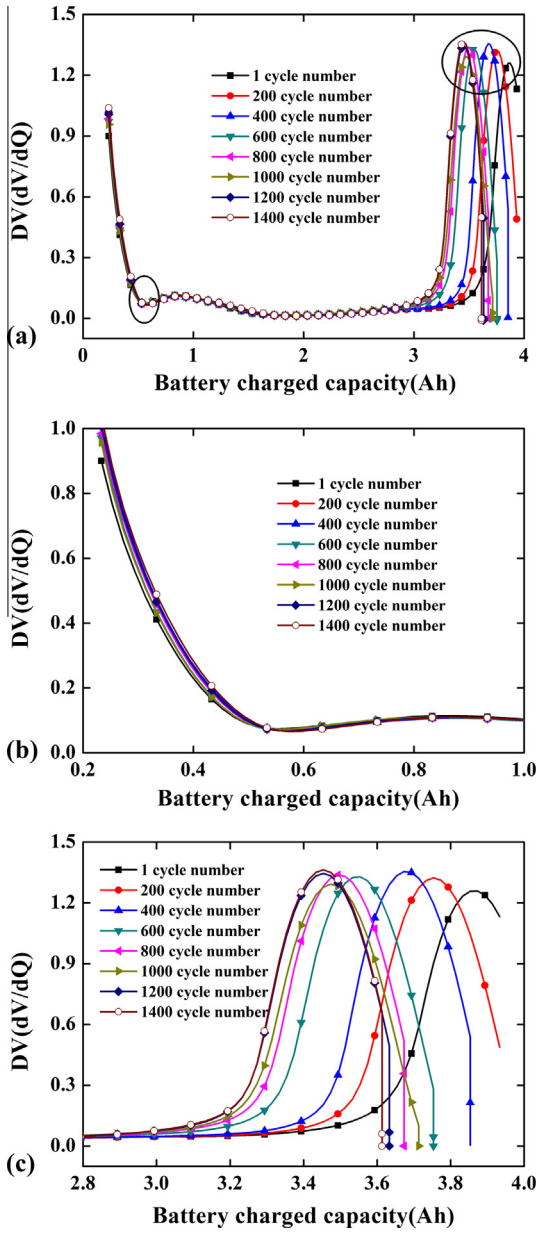


Fig. 6. DV curves with different time sizes of No. 1 battery cell calculated by the improved center least square method.

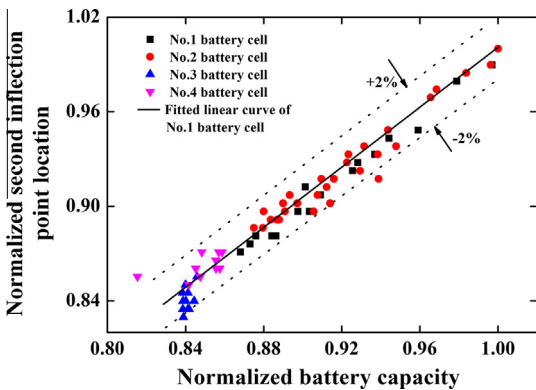


Fig. 7. Correlation between the normalized second inflection point location and the normalized battery cell capacity of the four battery cells.

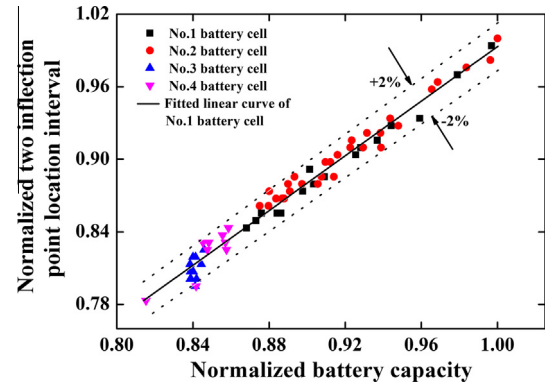


Fig. 8. Correlation between the normalized two inflection point location interval and the normalized battery cell capacity of the four battery cells.

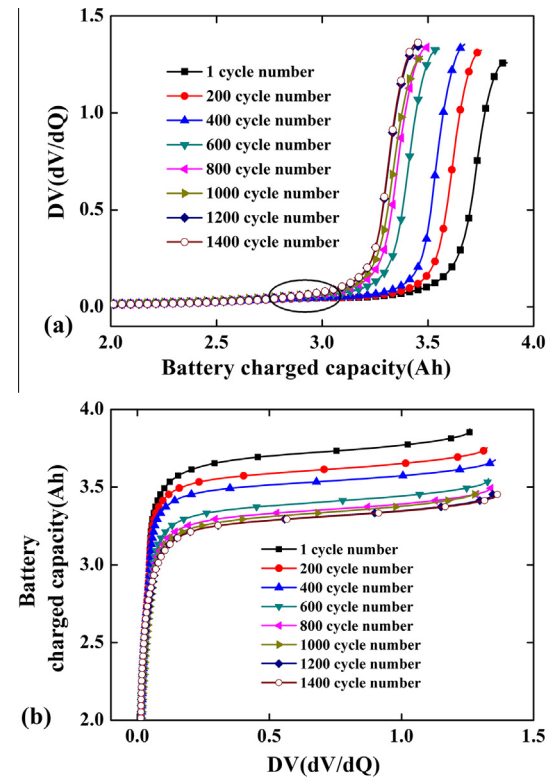


Fig. 9. (a) An enlarged view of parts of the $\frac{dV}{dQ} - Q$ curve shown in Fig. 6(a) and (b). Coordinate transformed $Q - \frac{dV}{dQ}$ curve.

where the parameter k_i presents the aging degree of a battery cell, Q_{0i} is the offset in the charged capacity direction.

As shown in Eq. (9), k_i represents the transformation parameter along the charged capacity direction, but this is not convenient for on-board identifying. Fig. 9(b) shows the $Q - \frac{dV}{dQ}$ curve, which is a coordinate transformed from the $\frac{dV}{dQ} - Q$ curve shown in Fig. 9(a). $h_i(Q_j^{-1})$ can be converted by the transformation function $F_{Q_i} = [k_i, h_{0i}]$ to be congruent with $h_0(Q_j^{-1})$, which is described as:

$$T = \sum_{j=1}^m (h_i(Q_j^{-1})/k_i + h_{0i} - h_0(Q_j^{-1}))^2 = T_{\min} \quad (10)$$

where $h_i(Q_j^{-1})$ is the $Q - \frac{dV}{dQ}$ curve of the No. i cycle number, $h_0(Q_j^{-1})$ represents the original $Q - \frac{dV}{dQ}$ curve which is provided by the manufacturer.

By solving $\frac{\partial T}{\partial k_i} = 0$, $\frac{\partial T}{\partial h_{oi}} = 0$ to minimize the solution of $T(k_i, h_{oi})$, k_i , h_{oi} is determined.

Fig. 10 shows the correlation between the transformation parameter k_i and the normalized battery cell capacity of the four

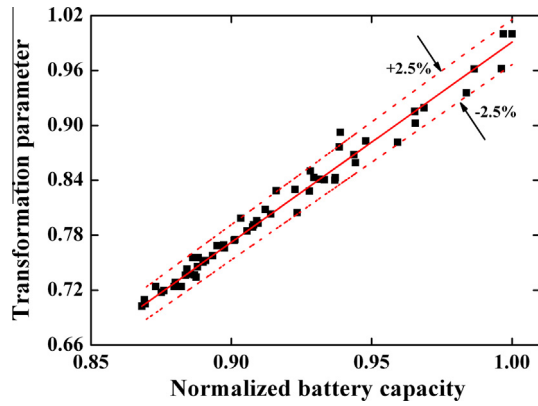


Fig. 10. Correlation between the transformation parameter k_i and the normalized battery cell capacity.

battery cells. The solid line in the figure gives the fitted linear relationship between the two values while the dotted line gives the corresponding error bound of $\pm 2.5\%$. A high relationship exists between k_i and the battery capacity, which confirms that the parameter k_i can also be used to indicate the battery capacity fading. The linear relationship of the battery capacity versus the transformation parameter k_i also should be calibrated off-line firstly. In actual application, the transformation parameter of the DV curve is calculated by Eq. (10). The SOH of the battery cell is then estimated using the established linear function.

6. Battery pack SOH monitoring framework

With the established linear correlation between the normalized battery capacity and the normalized location interval of the two inflection points or the transformation parameter, an on-board SOH monitoring framework is developed, as shown in Fig. 11. The measured V–Q curve is processed by an improved center least squares method to obtain the corresponding DV curve. When the DV curve reaches the first inflection point, the charged capacity calculated by the standard current integration method is calibrated

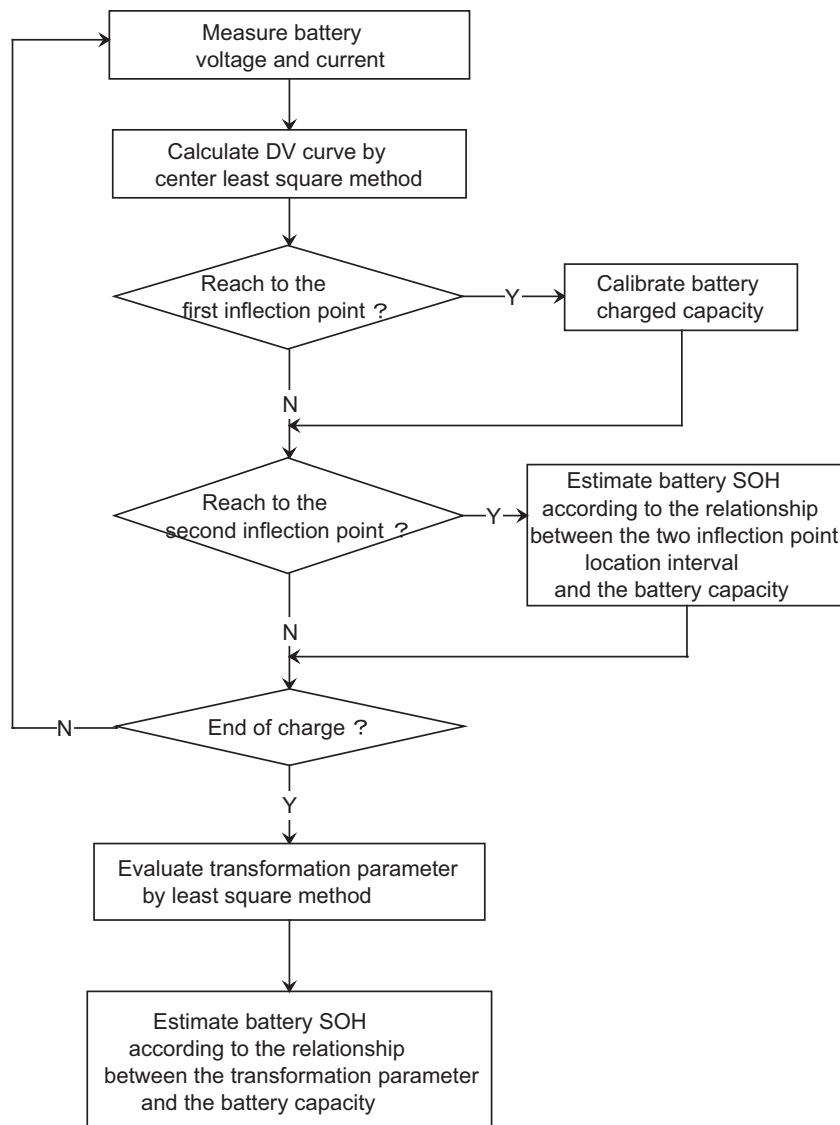


Fig. 11. On-board SOH monitoring framework.

to eliminate the cumulative error. If the DV curve reaches the second inflection point, the location interval of the two inflection points can be calculated. According to the established linear relationship between the battery capacity and the location interval of the two inflection points, the battery cell SOH is then estimated. If the DV curve of a battery cell cannot reach to the second inflection point when the end of charge occurs, the transformation parameter of the DV curve is calculated by Eq. (10). The SOH of the battery cell is then estimated using the transformation parameter. Moreover, the battery pack SOH can be evaluated with all the battery cell capacity being acquired.

7. Conclusion

A new algorithm to calculate the differential voltage (DV) curve is presented using an improved center least squares method. The basic principles of this algorithm are introduced, and the algorithm is used to obtain more robust and smoother DV curve compared to the numerical-derivative with smoothing method.

Given that the DV curve has two inflection points of interest, a new method is proposed to estimate a battery pack SOH based on the location interval of two inflection points or the transformation parameter of the DV curve. Four LiFePO₄ batteries in different aging states are tested to verify the validation of the proposed method. The results show that the linear regression of the location interval between two inflection points versus the battery capacity from one single battery cell is able to evaluate the SOH of the other three battery cells within 2% error bound. Transforming the measured DV curve to be consistent with the initial DV curve provided by the manufactory, the battery SOH is estimated according to the transformation parameter with the error being within 2.5%. This method can be applied to the battery cell which is unable to be charged up to the second inflection point at the end of charging process due to the battery cell inconsistencies in a battery pack. A battery pack SOH monitoring framework is inferred from the inflection point location and the transformation parameter of the DV curve to estimate the battery pack SOH.

Future investigations will focus on testing under different operating conditions and implementing the method within an actual BMS.

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